

An Introduction to Firefox OS

"We're out to make a difference, not a profit," says the project's home page. The Firefox operating system has come into the market with a big bang. Read about how to install it on your computer system.



Firefox OS (project name: *Boot to Gecko*, also known as *B2G*) is a Linux kernel-based open source operating system for smartphones and tablet computers. It is being developed by Mozilla, the non-profit organisation best known for its Firefox Web browser. The Firefox OS is about pushing the boundaries of what is possible with the Web on the mobile, and enabling entirely new segments of users to come online with their first smartphones.

The Firefox OS is designed to provide a complete community-based alternative system for mobile devices, using open standards and approaches such as HTML 5 applications, JavaScript, a robust privilege model, open Web APIs to communicate directly with cell phone hardware and an application marketplace.

Firefox OS terminology

There are a few terms that we need to get used to before we go further with the Firefox OS.

B2G: This is the short form for 'Boot 2 Gecko'. This is the main engine on which the Firefox OS runs and is the engineering code name for the Firefox OS. You will often see this term used to refer to the Firefox OS, since it was used for a long time before the project had an official name.

Firefox OS: Firefox OS is basically Mozilla's branding and support services applied on top of the B2G, to create a final release product.

Gaia: This is the user interface of the Firefox OS

platform. Gaia implements the lock screen, home screen and all the standard applications you expect on a modern smartphone. It is implemented entirely using HTML, CSS and JavaScript. Its only interfaces to the underlying operating system are through Web APIs that are implemented by the Gecko layer. Third party applications can be installed alongside the Gaia layer.

Gecko: This is the Firefox OS application runtime, i.e., this layer provides all the support for the triad of open standards—HTML, CSS and JavaScript. It makes sure these APIs work well on every operating system Gecko supports. This means that Gecko includes, among other things, a networking stack, graphics stack, layout engine, a JavaScript virtual machine and porting layers.

Gonk: Gonk is the lower level operating system of the Firefox OS platforms, consisting of a Linux kernel and user space hardware abstraction layer (HAL). The kernel and several of the user space libraries are common open source projects: Linux, libbus, Bluez, etc. Some of the parts of the HAL are shared with the AOSP (Android Open Source Project), such as GPS, camera and others. You could say that Gonk is a very simple Linux distribution. It is a porting target of Gecko, i.e., there's a port of Gecko to Gonk, just like there's a port of Gecko to OS-X, Windows and Android. Since the Firefox OS project has full control over Gonk, we can expose interfaces to Gecko that can't be exposed on other operating systems.